

Mystery Meat Analysis

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

[1]

Introduction

CITES was established in 1975 as a key legal framework to protect endangered species from global trade, yet from 1975 to 2014 the volume of CITES-listed organisms traded per year quadrupled to over 100 million whole-organism equivalents, a level which directly threatens both populations and whole ecosystems [2]. Rarity status attributed to a species often increases its price and demand, with critically endangered species often fetching the highest prices [3, 4]. The fallout of this is a positive feedback loop between rarity and demand which may drive an endangered species towards extinction [5]. Species-level identification of undocumented meat may uncover networks carrying out illegal imports of endangered species, and is therefore of great significance for the conservation efforts of these species.

If the morphology of a sample is compromised, DNA barcoding - comparing genetic sequences in a sample against known references - can aid in resolving the species of the sample [6]. Mitochondrial DNA (mtDNA) genes are commonly used for DNA barcoding because they occur in high copy numbers relative to nuclear DNA and typically lack recombination, keeping within-species diversity low and thus enabling species-level identification [7]. For example, cytochrome c oxidase I (COI), found within mtDNA, is the primary barcode marker in the animal kingdom, and reference sequences for COI are abundant in databases such as BOLD and NCBI which permit sample barcoding against a diverse set of organisms [8, 9].

This analysis aims to resolve the species of four meat samples intercepted by UK Border Force officers

at Heathrow Airport.

Methods

Pipeline Overview

Results

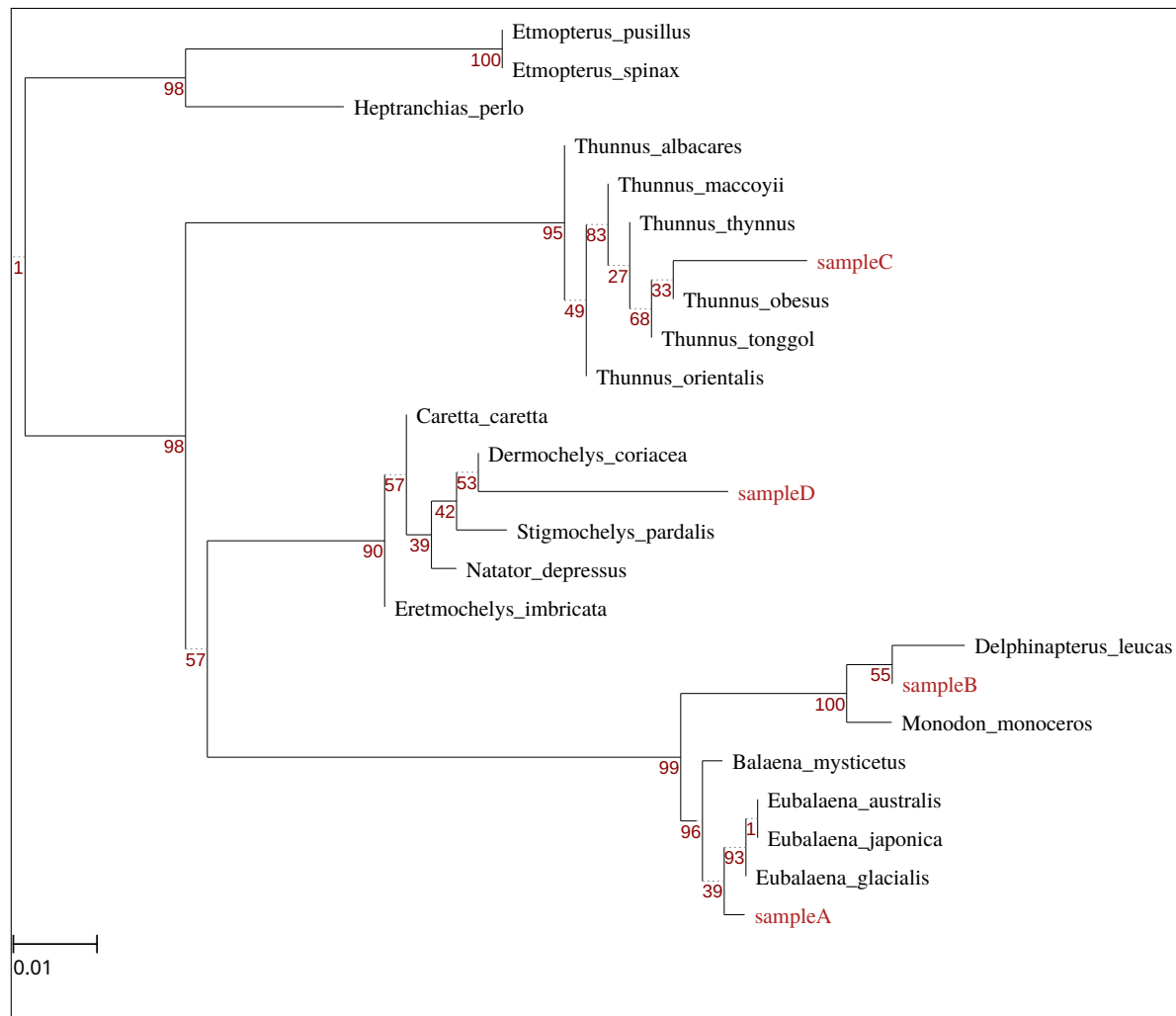


Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and top BLAST hits

Discussion

Where cyt b and D-Loop sequences do exist, very few of these have been generated as part of forensic studies [23], with the vast majority submitted to databases as part of phylogenetic, molecular ecological and conservation studies [24], [25], [26]. Such sequence data can be generated and accepted into databases without any need for standard protocols or quality control, raising doubts over their suitability for forensic application [7]

However, unlike BOLD, GenBank does not store sequence chromatograms, collection metadata or photographs [6]. BOLD is a curation tool that also stores sequences, while GenBank is just a sequence repository. [10]

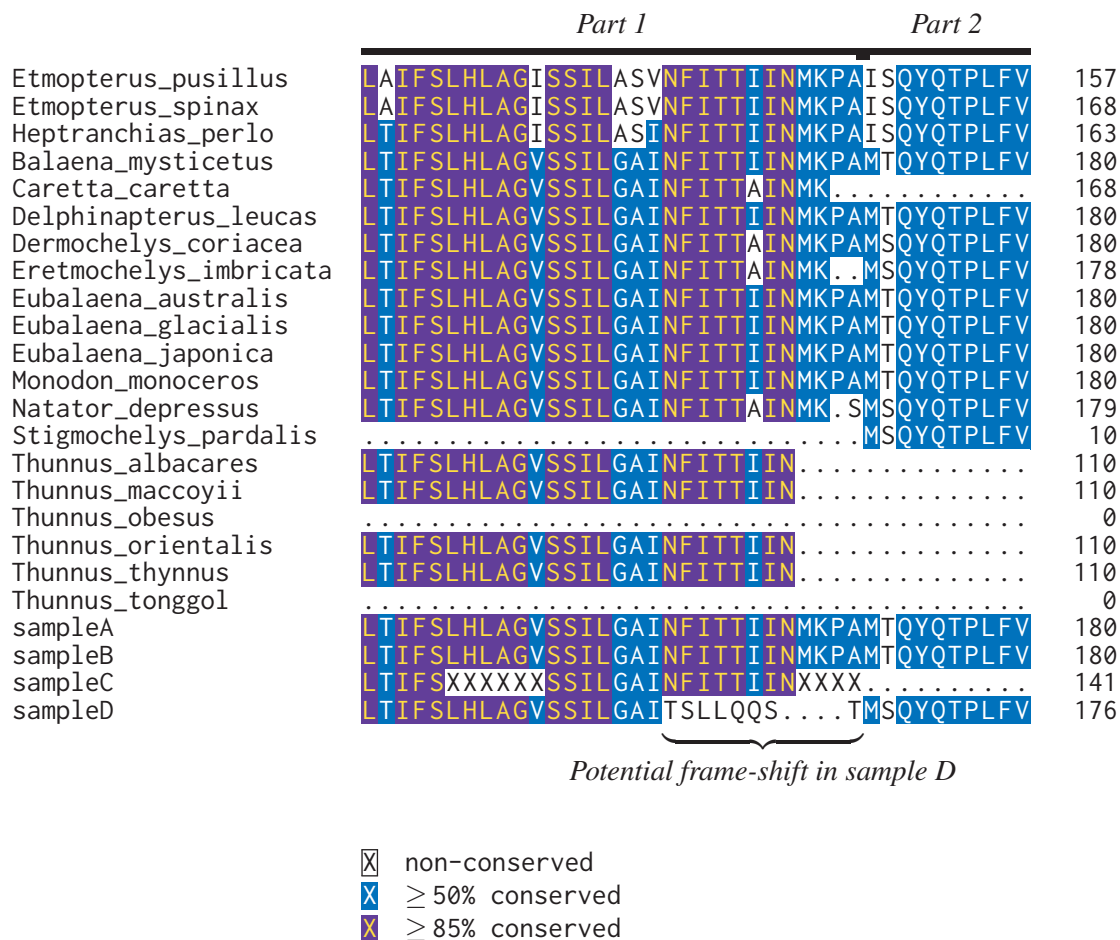


Figure 2: Window of supermatrix alignment with potential frame-shift error in sampleD near the highly-conserved NFITTIIN region.

Dermochelys_c	441	CACCTAGCTGGTGTTCATCAATTTAGGAGCTATTA	490
sampleD_part1	451	CACCTAGCTGGTGTTCATCAATTTAGGAGCTATT-AC	499

Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of sampleD_part1 against closest reference sequence (*Dermochelys coriacea*). The alignment gap indicates a possible base miss at position 486 in sampleD_part1 during sequencing.

References

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Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures for Manual Correction of Suspected Sequencing Error

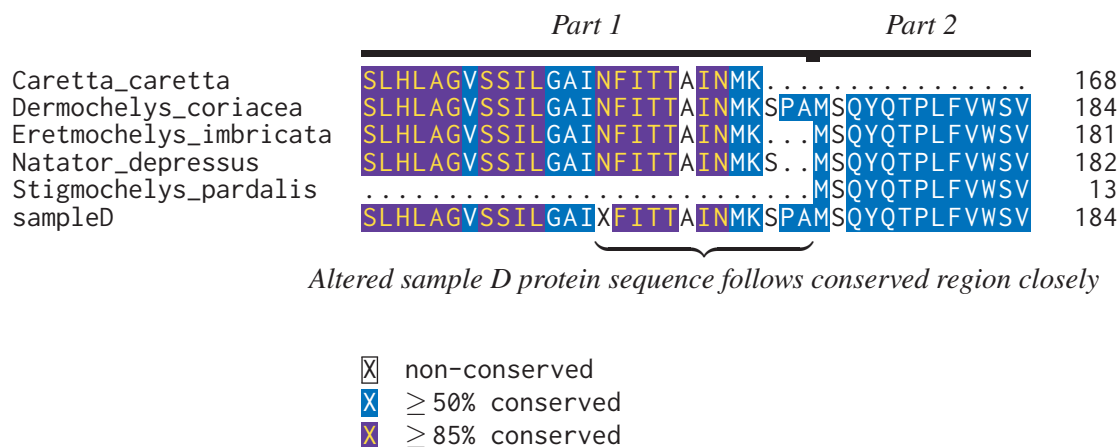


Figure 4: Subset of supermatrix alignment with sampleD and inner clade after inserting one ambiguous base (N) to the nucleotide sequence of sampleD_part1. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved NFITTIIN region compared to in Figure 2.

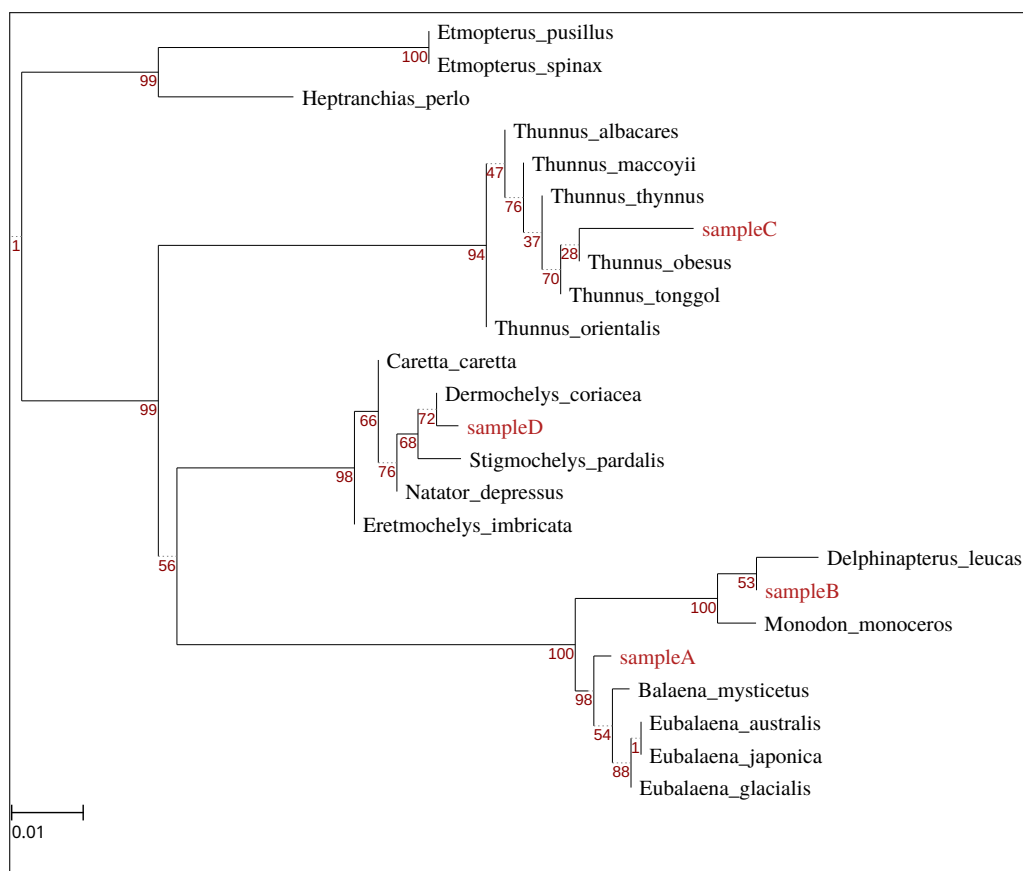


Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into sampleD_part1. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 1.